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Beyond Maximum Yield

The physics and dynamics of civilizational decay, societal collapse and polycrisis apocalypse



Hunter Hughes
Founder, H Heuristics

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The concepts. Maximum Civilizational Yield, Civilizational Excretion Deficit, post-peak civilization, polycrisis unravelling and Polycrisis Apocalypse are working concepts proposed by Hunter Hughes, Founder of H Heuristics, and are offered as hypotheses for testing.

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Data and methods. All data are public. Calculations, figures, simulations and any errors are the author's. Blue hyperlinks lead to the original sources; figure, table and section references link within the document.

EXECUTIVE SUMMARY

Executive Summary

Between 1950 and 2025 humanity enjoyed the fastest gains in its history. This report asks whether that was a peak, what a slow decay of the systems behind those gains would look like, and how it could unravel into a polycrisis.

Since 1950 world population has grown 3.2 times, income per person 5.0 times, energy use 6.3 times and plastic production about 230 times. Life expectancy has risen from 46 to 73 years (UN DESA, 2024; Bolt and van Zanden, 2024; Our World in Data, 2025a). This report calls the period from about 1950 to 2025 an era of **Maximum Civilizational Yield**: the time when the net gains from industrial civilization, after subtracting the costs of producing and maintaining them, were probably at their highest. It then asks what comes next. The question is not whether output can keep growing. It is whether the *systems that turn output into lasting human well-being*, such as ecosystems, infrastructure, institutions and trust, are being worn down faster than they are repaired.

3.1%
yearly growth of world income per person in the 1960s, against 1.2% in 2019–22; the 2000s bounce came from China and India catching up

356 Gt
fossil CO_2 emitted in the 2010s, three times the 117 Gt of the 1960s

32%
Western share of world output (PPP) in 2024, down from 46% in 1990, as its population share fell to 10%

43%
of 400 stylised runs on a prolonged-polycrisis path cross the unravelling threshold by 2100, against 0% on a managed path and 95% on a severe one

Principal findings

- **The 1950–2025 surge was real and unusual.** Income per person grew 3.1% a year in the 1960s, the fastest decade since 1950, and cereal yields tripled (Figure 2). It rested partly on one-off windfalls: cheap high-quality fuel, unused farmland and a young, growing workforce (Section 3).
- **The rate of improvement has slowed while the bill has grown.** Gains in income, life expectancy, crop yields and energy growth were all strongest in the 1960s or 1970s, whereas the fossil CO_2 added each decade has risen every decade, from 117 to 356 Gt. A rough index of gain per unit of added CO_2 fell to about a seventh of its 1960s level (Figure 5). This supports, but does not prove, a post-peak reading (Section 4).
- **Civilization has a metabolism and a waste problem.** It takes in about 632 exajoules of energy and 106 billion tonnes of materials a year, and returns only about 7% of those materials to use. The atmosphere holds 427 ppm of CO_2, up from 316 in 1959 (Figure 6). The excretion parallel is a useful warning about stocks and flows, not a literal law.

- **Decay is a maintenance story.** Materials science shows that damage from repeated shocks adds up, and that deferred repair can turn a survivable load into a failure (Figure 7). Societies pay a rising price for complexity, with falling marginal returns (Tainter, 1988; Bloom et al., 2020) (Section 5).
- **The West is the most complex system and the oldest.** Its income growth per person fell from 3.5% a year in the 1960s to 1.4% in the 2010s, its fertility is 1.5 children per woman, government debt in the US, UK and France has roughly doubled since 2007, and trust in government is low. It is still rich, healthy and resilient. The report sets out what relative decline, institutional decay and systemic collapse would each look like, and why only the first is clearly under way (Section 6).
- **Abrupt change grows out of gradual stress.** Systems with feedback show tipping points, hysteresis and early-warning signals, and tightly coupled networks fail in cascades rather than slopes (Figure 10, Figure 11). Warm-water coral reefs have already passed their thermal tipping point (Lenton et al., 2025) (Section 7).
- **Polycrisis unravelling is a loop-gain problem.** If climate, pollution, poverty and conflict each pass on enough of their stress to the others, any shock is amplified by 1 ÷ (1 − gain), and above a gain of 1 the loop runs away (Figure 12). Capacity to absorb stress is what keeps the gain below 1 (Section 8).
- **A Polycrisis Apocalypse is a tail scenario, not a forecast.** In a stylised model it needs both rising pressure and eroding capacity. Managed transition avoids it, a prolonged polycrisis risks it in a minority of runs, and a severe path makes it likely (Figure 13). Some legacies, such as CO_2 and sea level, outlast any scenario (Section 9).
- **Resilience is the countervailing lever.** Reducing the pressure, rebuilding the buffers and cutting the coupling all lower the loop gain. The report ends with six falsifiable tests that would show whether the post-peak thesis is right (Section 10).

How to read this report

Sections 1–3 set out the thesis, the physical toolkit and the evidence for a 1950–2025 peak. **Sections 4–5** cover diminishing returns, the metabolic bill and the slow decay of systems. **Section 6** applies the ideas to the Western world, the central case. **Sections 7–9** follow decay into tipping points, polycrisis unravelling and long-run scenarios. **Section 10** turns to resilience and a test programme. Everything measured comes from public data listed in the references. Models are labelled as stylised, and the report separates *observed trends*, *mechanisms* and *scenarios*.

READ FIRST

Epistemic status

The thesis is a hypothesis. This report does not claim that a global maximum has been passed, that Western societies will collapse, or that a Polycrisis Apocalypse is likely. It claims that the evidence is consistent with a post-peak reading, that the physical and mathematical parallels are useful for asking better questions, and that the downside tail deserves the same planning attention as other low-probability, high-consequence risks.

SECTION 01

The Question: Yield, Decay and Collapse

Collapse may not begin when a civilization runs out of productive capacity. It may begin when the systems that turn that capacity into sustainable human development wear out.

IN BRIEF

- Output can keep rising while the ability to sustain the conditions behind it falls.
- Six stages link a peak in net yield to decay, collapse, polycrisis unravelling and, in the extreme, a Polycrisis Apocalypse.
- Each stage has an indicator, so the thesis can be tested rather than only told as a story.

1.1 The central proposition

Most debates about collapse ask whether we will run out of something: oil, soil, water, minerals. This report asks a different question. A civilization is a machine for turning energy and materials into human well-being, and it needs a second set of systems to keep that machine running: stable ecosystems, working infrastructure, capable institutions and social trust. The proposition is that **civilizational collapse is not necessarily initiated by the exhaustion of productive capacity, but by the progressive deterioration of the systems required to convert productive capacity into sustainable human development**.

A society can therefore keep producing more goods, energy, technology and infrastructure while becoming less able to maintain the ecological, institutional and social conditions those outputs depend on. This is the difference between *gross* yield, which is what we usually measure, and *net* yield, which is what is left after the costs of making and keeping the gains are deducted.

1.2 Six stages from peak to apocalypse

The argument is built as a ladder of six stages, each defined so that it can be measured. Table 1 gives working definitions and the evidence status as of 2026. The stages are not a destiny, and different regions may sit at different rungs at the same time. Figure 1 sketches the idea: a rise to a peak in net yield, then three possible futures.

1.3 What this report is, and is not

It is a **framework paper**: it assembles public data, established physical ideas and simple models to ask whether the evidence is consistent with a post-peak, decay and unravelling story. It separates three kinds of claim. **Observed trends** come from data, **mechanisms** are the physical and mathematical ideas that could explain them, and **scenarios** are what-if projections.

It is not a prediction that civilization will end. Historians stress that collapse is usually a *simplification*, a loss of scale and complexity over decades or centuries that is uneven across places and groups (Tainter, 1988; Scott, 2017; Cline, 2014). Collapse here means failure of essential functions, not human extinction.

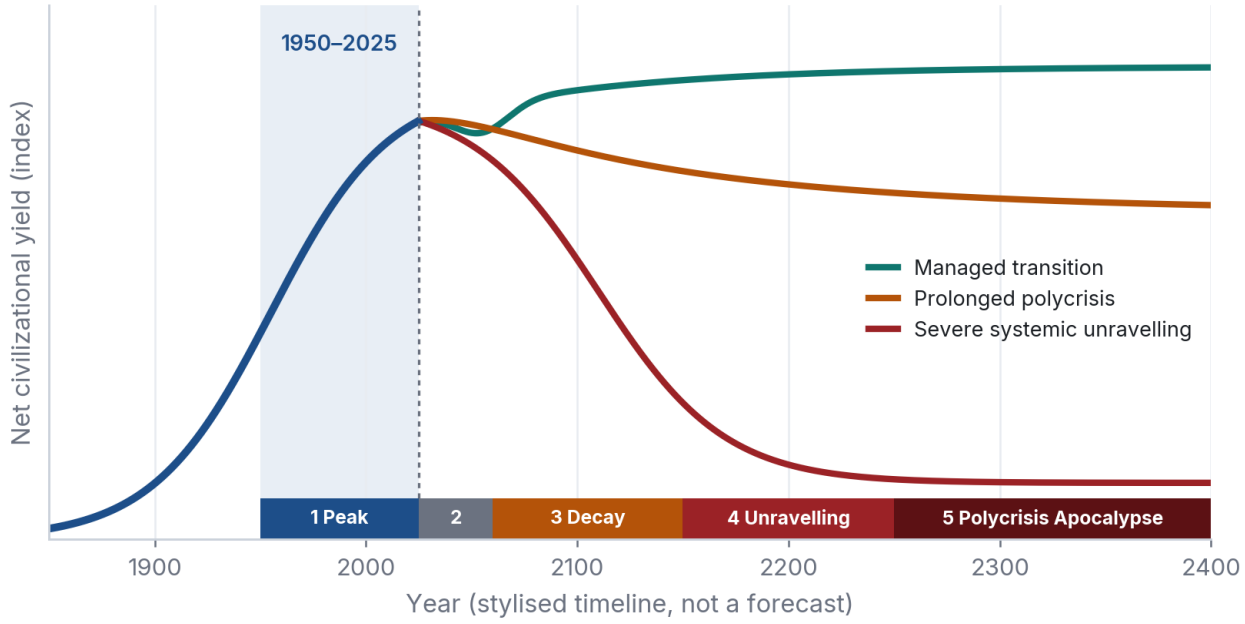
Table 1 Six stages, their meaning, how to measure them and where the evidence stands

Stage	Working definition	Main indicator	Status in 2026
1 Maximum Civilizational Yield	Peak of net developmental returns after deducting the costs and liabilities of producing them	Rates of gain per unit of throughput and liability	Probably 1950–2025; mixed by indicator
2 Post-peak civilization	Marginal gains fall because of resource, ecological, institutional and maintenance costs	Falling growth and productivity; rising maintenance share	Visible in growth, yields, research
3 Civilizational decay	Accumulating damage, deferred maintenance and falling adaptive capacity	Infrastructure condition, debt, trust, buffers	Partly observed in some Western systems
4 Societal and systemic collapse	Failure of essential functions, uneven across regions and sectors	Loss of power, water, food, order, public services	Regional and local only
5 Polycrisis unravelling	Linked crises reinforce one another as weakened systems stop absorbing shocks	Loop gain, coincident hazards, cascades	Early signs; no global runaway observed
6 Polycrisis Apocalypse	Persistent, widespread loss of survivability, services and ecological conditions	Sustained fall in life expectancy and food, water, energy access	Not observed; a tail scenario

Author's definitions, consistent with the research overview that frames this report (Hughes, 2026a). Status column is the author's reading of the evidence in Sections 3–8.

Figure 1 A stylised path: a rise to peak net yield, then three possible futures

Net civilizational yield (index) from 1850 to 2400 with the five stages marked along the base; stage 6 is the persistent low state at the right



Source: author's schematic. Not a forecast; the curve shows the shape of the argument, not measured values.

Three tests organise the evidence: whether net gains per unit of cost have fallen (the **peak test**), whether liabilities build up faster than they are cleared (**accumulation**) and whether coupling has tightened and buffers thinned (**fragility**).

SECTION 02

The Physics Toolkit: Five Foundations

Five ideas from physics, materials science and mathematics give precise vocabulary for how a complex society can lose capacity without running out of resources.

IN BRIEF

- Thermodynamics, metabolism, fatigue, nonlinear dynamics and network science each explain one piece of decay.
- The sun supplies far more energy than humanity uses, so the entropy law is not a ceiling on growth; waste, materials and maintenance are the binding costs.
- Every analogy has limits, and this section states them.

2.1 Five interacting foundations

The thesis rests on five foundations, summarised in Table 2. Each is an established field with its own equations and evidence. None says that societies must collapse. Together they say something more modest and more useful: that complex systems have *running costs*, produce *residuals*, *accumulate damage*, can *switch abruptly* and are *connected in ways that spread failure*.

Table 2 The theoretical architecture: five explanatory foundations

Foundation	Physical or scientific parallel	What it helps explain
Thermodynamics	Energy dissipation and entropy production	The physical costs of maintaining complex systems
Metabolism and excretion	Resource intake, transformation, residual outputs	Waste accumulation and environmental burdens
Materials science	Fatigue and cumulative damage	How repeated shocks weaken infrastructure and institutions
Nonlinear dynamics	Bifurcations, tipping points, hysteresis	Why gradual deterioration can produce abrupt transitions
Network science	Cascading failure and percolation	How interconnected crises propagate through society

Author's framework.

2.2 Thermodynamics: civilization as an engine that must dump heat and waste

A living cell, a city and a civilization have something in common: each is an ordered structure held far from equilibrium by a constant flow of energy. Schrödinger described life as feeding on "negative entropy", and Prigogine showed that such *dissipative structures* stay organised only while energy flows through them and disorder is exported (Schrödinger, 1944; Prigogine and Stengers, 1984). Georgescu-Roegen applied the same logic to economies, arguing that every production process turns usable energy and concentrated materials into low-grade heat and dispersed waste (Georgescu-Roegen, 1971). Humanity now runs on about 20 terawatts, roughly 2.5 kilowatts for every person, about 25 times the 100 watts of a human body (Our World in Data, 2025a).

This does **not** imply a hard thermodynamic limit. The Earth is an open system: it intercepts about 170,000 terawatts of sunlight, so human power use is roughly 0.01% of the flow. The second law forbids perpetual motion, not growth. What it does imply is that *every* increase in order has a running cost, that waste has to go somewhere, and that a system whose waste output exceeds the rate at which its surroundings can absorb it will accumulate that waste as a stock. The binding limits are therefore practical: the energy return on the fuels we use, the supply of concentrated materials, and the sinks for heat, carbon and pollutants. Section 4 returns to each.

2.3 Metabolism and excretion: what goes in must come out

Biologists measure organisms by their *metabolism*, the rate of taking in energy and materials, and their *excretion*, the handling of waste. An organism that cannot clear its waste poisons itself even when food is plentiful. A civilization is not an organism, but the bookkeeping is the same. Section 4 introduces a Civilizational Excretion Deficit, a condition in which residual outputs persistently exceed the combined capacity of natural assimilation, recycling, treatment and clean-up. The parallel is deliberately kept brief and is used only for stock-and-flow accounting.

2.4 Materials fatigue: damage that adds up

Engineers know that a bridge does not fail only when a single load exceeds its strength. It can fail after many loads, each below that strength, because tiny cracks grow with every cycle. Miner's rule adds up fractional damage from each load, and the S–N curve shows that doubling the size of a load can cut the number of cycles to failure by a factor of around nine. The idea carries naturally to societies: repeated floods, heatwaves, financial crises and pandemics each take a bite out of buffers, and recovery that is deferred leaves the next shock to land on a weaker base. Section 5 shows this with a simple simulation.

2.5 Nonlinear dynamics and networks: why decay can turn abrupt

In a linear system, twice the stress produces twice the strain. Real systems with feedback are often not linear. They can have two stable states, such as a clear lake and an algae-choked one, and a tipping point between them. Once past it, going back needs stress to fall far below the level that caused the shift, a property called *hysteresis* (Scheffer et al., 2001). Close to a tipping point, recovery from small shocks slows down, a signature known as critical slowing down that offers an early warning (Scheffer et al., 2009). When many such systems are linked in networks, failures can cascade: in coupled networks, a small initial failure can trigger a collapse that no single network would suffer (Buldyrev et al., 2010; Helbing, 2013). Section 7 develops both ideas.

2.6 Where the analogies break

Physical parallels clarify, but they can also mislead. Societies have foresight and can change their own rules, which atoms and bridges cannot. Table 3 lists the main strength and the main limit of each foundation, so that the rest of the report does not lean on any of them further than it can bear.

Table 3 What each analogy explains well, and where it should not be pushed

Foundation	Where it is strongest	Where the analogy breaks
Thermodynamics	Energy and waste are real, unavoidable costs of order	The sun supplies far more energy than we use, so entropy alone sets no near-term ceiling
Metabolism	Stocks build up when inflow exceeds outflow (CO_2, plastics)	Societies can build new organs: recycling, treatment, carbon removal
Materials fatigue	Deferred maintenance turns survivable loads into failures	Societies can redesign structures and rules, not just repair them
Nonlinear dynamics	Thresholds and hysteresis are well documented in ecosystems and climate	Social thresholds are harder to measure and may be crossed by choice or reversed by policy
Network science	Dependencies between grids, supply chains and finance can spread failure	Real networks have firebreaks, redundancy and human responses that models omit

Author's assessment.

SECTION 03

Maximum Civilizational Yield, 1950–2025

Never before had so many people become so much healthier, richer and better fed in so short a time. That is also why 1950–2025 is the natural candidate for a peak.

IN BRIEF

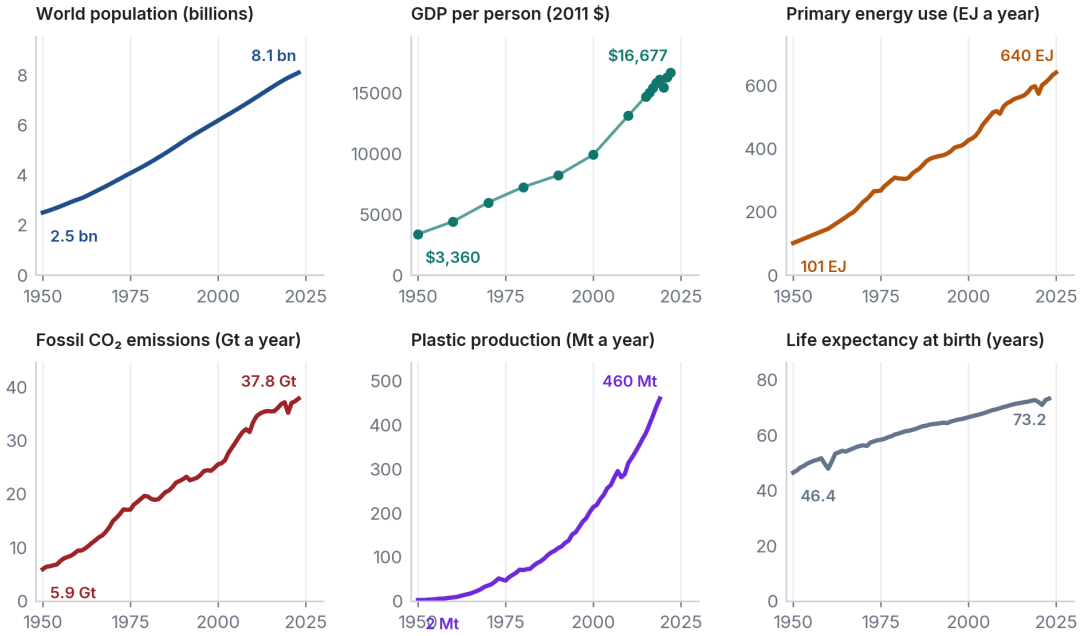
- World income per person rose 5.0 times and life expectancy by about 27 years.
- The same period multiplied energy use, emissions and plastic output several times over.
- Part of the gain came from one-off windfalls that cannot be repeated at the same scale.

3.1 Why 1950

Historians date a "Great Acceleration" to the middle of the twentieth century. Across twenty-four indicators of human activity and Earth-system change, Steffen and colleagues found that almost every curve, from population and GDP to fertiliser use, carbon dioxide and ocean acidification, bends upward after 1950 (Steffen et al., 2015). The postwar settlement brought peace among major powers, expanding trade, mass vaccination, cheap oil and the Green Revolution. Figure 2 shows six of these curves.

Figure 2 Between 1950 and today, people, output, energy, emissions and plastics all multiplied, and life expectancy rose by 27 years

Six indicators of the Great Acceleration, 1950–2025 (latest available year)



Source: UN World Population Prospects 2024, Maddison Project Database, Energy Institute via Our World in Data, Global Carbon Project, Geyer et al. and OECD, (UN DESA, 2024; Bolt and van Zanden, 2024; Our World in Data, 2025a; Our World in Data, 2025; Geyer et al., 2017). Population and life expectancy end in 2023; plastics in 2019; GDP in 2022.

3.2 What was gained

The gains were large by any measure. Income per person in constant dollars rose from about $3,360 to $16,677, even as the number of people grew 3.2 times, so total world output grew roughly 16-fold (Bolt and van Zanden, 2024). Life expectancy at birth rose from 46 to 73 years. Deaths of children under five fell from 94 to 37 per 1,000 live births between 1990 and 2024 (World Bank, 2026). Cereal yields rose from 1.4 to 4.2 tonnes per hectare, which is why cereal output could keep pace with a population that has more than tripled (FAO, 2025).

Table 4 The yield and its costs, 1950 to the latest year

Indicator	1950	Latest	Multiple	Source
World population (billion)	2.5	8.1 (2023)	×3.2	UN WPP
GDP per person (2011 $)	3,360	16,677 (2022)	×5.0	Maddison
Life expectancy at birth (years)	46.4	73.2 (2023)	+26.8 yrs	UN WPP
Primary energy (EJ a year)	101	632 (2024)	×6.3	Energy Institute
Fossil CO_2 (Gt a year)	5.9	37.8 (2023)	×6.4	Global Carbon Project
Plastic production (Mt a year)	2	460 (2019)	×230	OECD, Geyer et al.

(UN DESA, 2024; Bolt and van Zanden, 2024; FAO, 2025; Our World in Data, 2025a; Our World in Data, 2025; Geyer et al., 2017; NOAA, 2026). Calculations by the author.

The right-hand rows are the other half of the ledger. The same period that produced the gains multiplied the throughput of energy, materials and waste, which is the subject of Section 4.

3.3 Why the yield was unusually high

Four conditions made the postwar decades exceptional, and each was at least partly a windfall. First, **energy was cheap and concentrated**. Oil, gas and coal came from fields with very high energy returns. Second, **technology had low-hanging fruit**: electrification, antibiotics, synthetic fertiliser, high-yielding crop varieties and jet transport were each first-order breakthroughs that could be rolled out worldwide. Third, **demography helped**: falling child mortality followed by falling fertility produced a bulge of working-age adults. Fourth, **there were unused stocks** of fertile soil, groundwater, forests and ocean fish to draw on, and an atmosphere that seemed able to absorb emissions at no visible cost.

None of these is repeated on the same terms: the easiest energy has been found, the biggest technology gains captured, the demographic dividend is ending in most rich and many middle-income countries, and the stocks have been drawn down. The point is not that progress stops, but that the same effort now buys less, as Section 4 shows. The gains were also uneven: the West held 46% of world output in 1990 and 32% in 2024 while China rose from 3% to 19%, so a world peak can hide a plateau in some regions and catch-up in others.

SECTION 04

Past the Peak: Diminishing Returns and the Metabolic Bill

The best decade for most of the gains was the 1960s or 1970s. Since then each unit of growth has come with a larger bill in emissions, waste and maintenance.

IN BRIEF

- Growth in income, crop yields, energy use and life expectancy was fastest decades ago.
- Research effort needed for each new advance has risen sharply, and the energy cost of getting energy matters more than the headline supply.
- Residual outputs, from CO_2 to plastic, accumulate faster than they are cleared, a brief parallel to excretion.

4.1 Is the rate of gain falling?

The simplest test of a post-peak world is to compare decades. Figure 3 shows four rates of gain. World income per person grew 3.1% a year in the 1960s, the best decade, and only 1.7% in the 2010s, 1.2% in 2019–22. Cereal yields grew 3.0% a year in the 1960s and about 1.5% in the 2010s. Energy use, which grew 4.7% a year in the 1960s, grew 0.7% in the 2010s. Life expectancy gained 8.4 years in the 1960s, partly a rebound from the 1959–61 famine in China, and 1.8 years in the 2010s (Bolt and van Zanden, 2024; UN DESA, 2024; FAO, 2025; Our World in Data, 2025a).

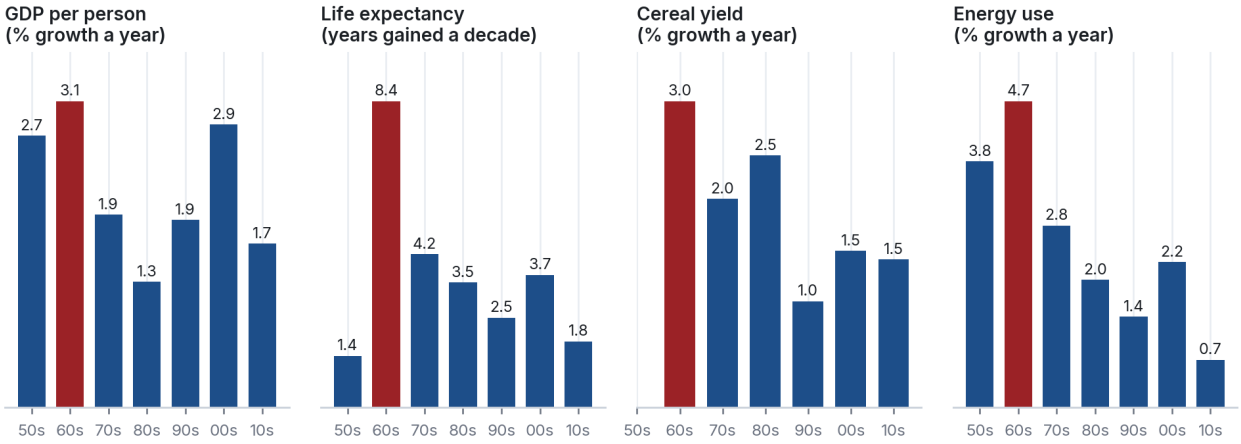
Two cautions apply. First, the picture is not a smooth decline. World growth per person rebounded to 2.9% a year in the 2000s, because China and India grew fast and joined the world economy; that is a one-time catch-up, not a new trend. Second, slower growth from a higher base is partly arithmetic. A rise of 1.7% on today's income adds more dollars than 3.1% did in 1960. The right reading is that the *rate* of improvement has slowed, while the *level* remains high and still rising slowly.

4.2 Ideas are harder to find

A second sign is the cost of progress. Bloom and co-authors measured research productivity across fields and found that it is falling fast. The number of researchers needed to double the density of computer chips is more than 18 times larger than in the early 1970s, and across the whole US economy, research productivity has fallen by a factor of 41 since the 1930s, or more than 5% a year on average (Bloom et al., 2020). Growth has held up only because the number of researchers has multiplied. This is a diminishing-returns story in its purest form: the same hours of research buy fewer new ideas, so keeping the growth rate steady needs ever more people and money.

Figure 3 The fastest decade for each gain was the 1960s or 1970s; none has matched it since

Four rates of gain by decade, 1950s to 2010s



Decade starting 1950 (50s) to 2010 (10s). Peak decade in red. Cereal yield starts in the 1960s; life expectancy in the 1960s rebounds from the 1959–61 famine.

Source: Maddison Project Database, UN WPP 2024, FAO, Energy Institute via Our World in Data; author's calculations (Bolt and van Zanden, 2024; UN DESA, 2024; FAO, 2025; Our World in Data, 2025a). The final bars cover 2010–2020 for GDP and life expectancy; cereal and energy growth for 2014–24 is 0.8% and 1.2%.

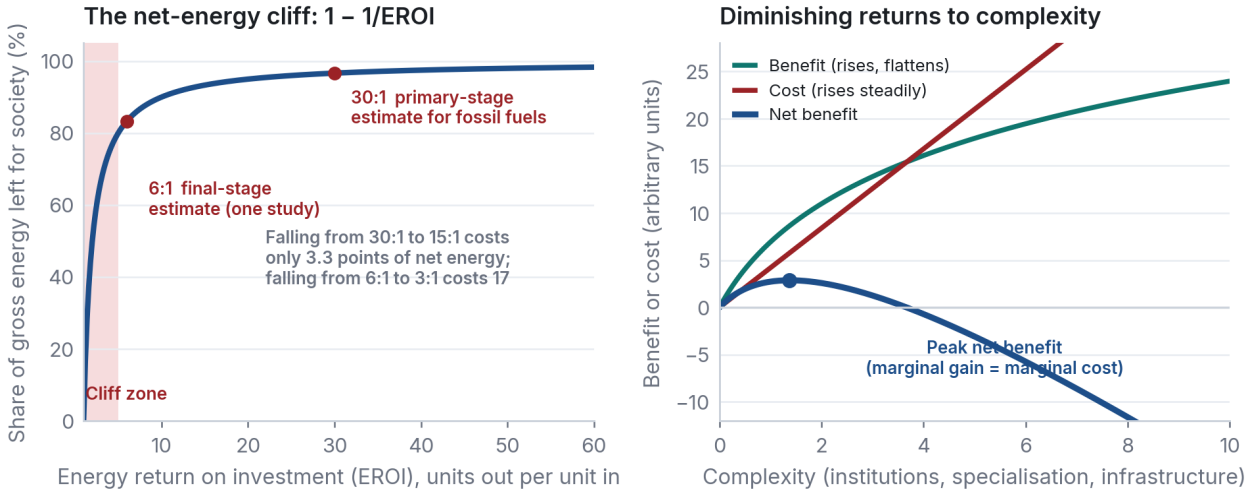
4.3 Energy: the cost of getting energy

Energy supply is not what matters most. What matters is the *net* energy left after the energy spent on finding, extracting, refining and delivering it. The ratio of energy out to energy in is the energy return on investment, or EROI. If EROI is 30:1, one thirtieth of the energy produced is spent producing it. If it is 3:1, a third is. The relationship between EROI and the share left for society is 1 − 1/EROI, which is almost flat at high returns and steep at low ones, the so-called net-energy cliff (Figure 4, left panel).

Estimates of current EROI are debated. Hall and colleagues argued that EROI for oil and gas has fallen over the past century as the best fields were used first (Hall et al., 2014). Brockway and colleagues agree that primary-stage returns for fossil fuels are about 30:1, but when the measure is taken where energy enters the economy, as electricity or petrol, they find about 6:1 and falling (Brockway et al., 2019). Other researchers dispute the method, and renewables have very different cost structures. A study of 'energy and quality of life' found that well-being measures rise with energy availability and flatten at higher levels, suggesting a threshold in the tens-to-one (Lambert et al., 2014). The sound conclusion is limited: a long era of very high energy returns has probably passed, and the clean transition is partly a bet that new sources can restore a high enough net return at lower environmental cost.

Figure 4 **Net energy falls slowly until returns get low, then fast; likewise, benefits of complexity flatten while costs keep rising**

Left: the net-energy cliff. Right: diminishing returns to complexity (Section 5)



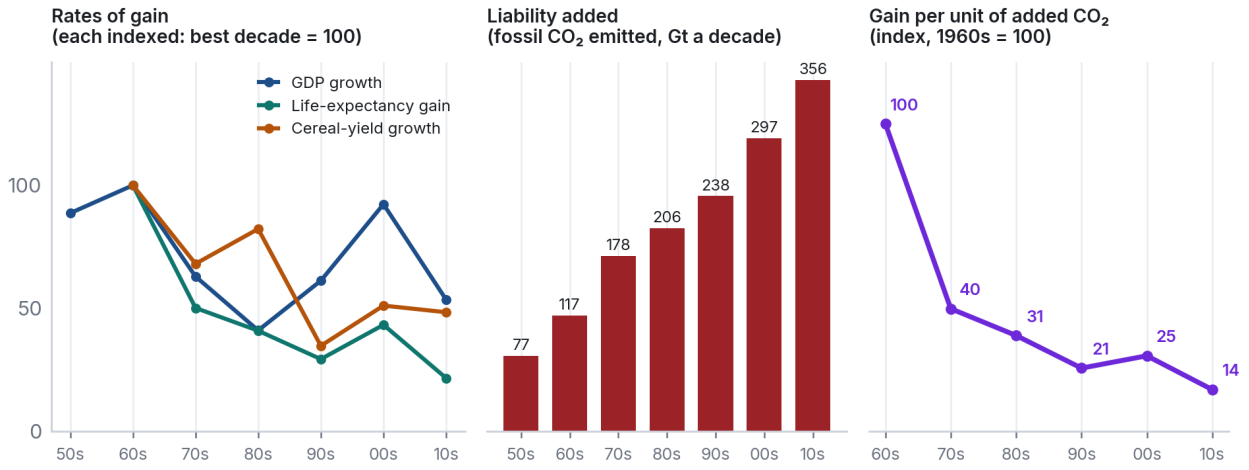
Source: author's calculations. Left: net share = 1 – 1/EROI, with 30:1 and 6:1 from (Brockway et al., 2019). Right: stylised, benefit = 10·ln(1 + c) and cost = 4.2·c.

4.4 The yield–liability ledger

To compare gains with costs, Figure 5 puts the three rates of gain on one scale (each indexed so its best decade is 100), shows the fossil CO_2 emitted in each decade, and divides one by the other. The result is a diagnostic, not a measurement of cause: GDP growth is not caused by CO_2 alone, and efficiency improvements have lowered the carbon in each dollar. Even so, the ratio is informative. The average gain index fell from 100 in the 1960s to about 41 in the 2010s, while CO_2 added per decade rose from 117 to 356 Gt. The gain per unit of added CO_2 fell to about one seventh of its 1960s level. If CO_2 is taken as a proxy for the wider growth in liabilities, the peak test is passed: the yield per unit of liability peaked early and has fallen since.

Figure 5 **Gains per decade have fallen from their peak while liabilities added per decade have risen every decade**

Rates of gain, CO_2 emitted per decade, and the ratio of the two



Source: author's calculations from (Bolt and van Zanden, 2024; UN DESA, 2024; FAO, 2025; Our World in Data, 2025). Cereal-yield index begins in the 1960s. Ratio = mean of the three indexed gains ÷ CO_2 emitted in the decade, 1960s = 100. A diagnostic, not a causal estimate.

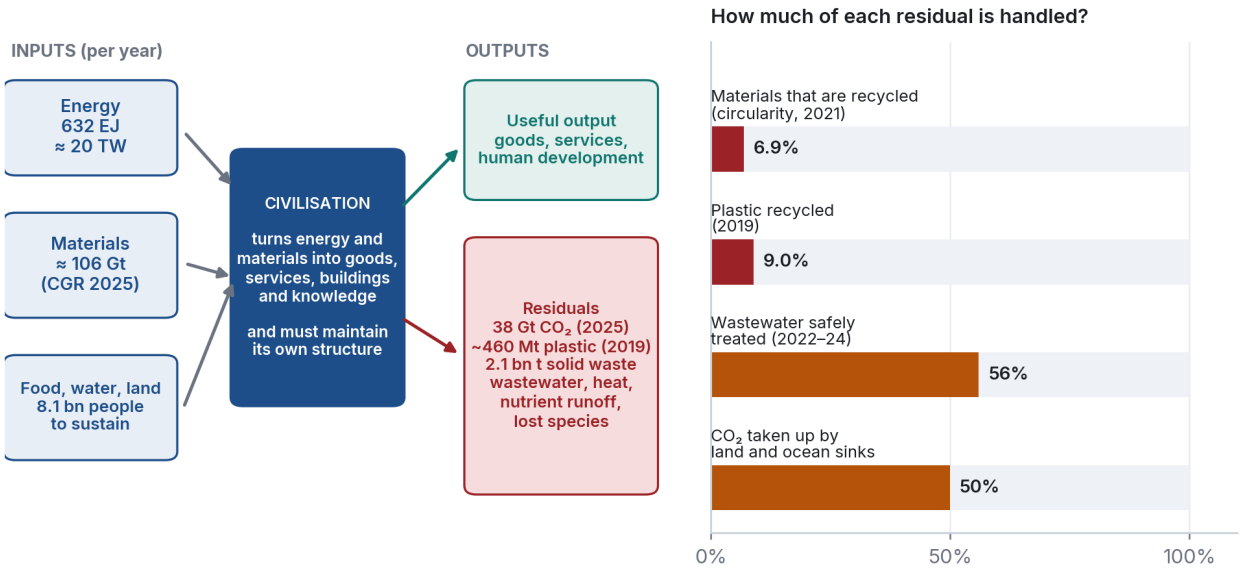
4.5 Civilization's metabolism and the excretion deficit

The parallel to metabolism is brief but useful. Civilization takes in about 632 exajoules of energy and 106 billion tonnes of materials a year, builds goods, infrastructure and knowledge, and returns residuals: heat, carbon dioxide, plastics, solid waste, wastewater, fertiliser runoff and lost habitat (Figure 6, left). Organisms have specialised organs for excretion, such as kidneys and liver. Civilization has none: waste handling is spread across industries, households, ecosystems and trade, and is often under-priced.

This report defines a **Civilizational Excretion Deficit** as a state in which the generation of residual burdens persistently exceeds the combined capacity of environmental assimilation, recycling, treatment and remediation. The evidence for a deficit is mixed by residual, as Figure 6 (right) shows. The global economy returns only 6.9% of its material inputs to use, down from 9.1% in 2018 (Circle Economy, 2025). Of about 460 million tonnes of plastic produced in 2019, around 9% was recycled and about 22% mismanaged (OECD, 2022). Roughly 56–58% of household wastewater is safely treated (UN-Water, 2024). Municipal solid waste is projected to grow from 2.1 billion tonnes in 2023 to 3.8 billion by 2050 (UNEP and ISWA, 2024). Air pollution is linked to 7.9 million deaths a year (Health Effects Institute, 2025). Carbon dioxide is the clearest case: land and ocean sinks absorb about half of emissions, leaving the rest to accumulate, so atmospheric CO_2 rose from 316 ppm in 1959 to 427 ppm in 2025 and fossil emissions reached a record 38.1 Gt in 2025 (Friedlingstein et al., 2025; NOAA, 2026).

Figure 6 Civilization's inputs and outputs: most residual flows are handled only in part

Left: yearly inputs and outputs. Right: share of each residual that is recycled, treated or absorbed



Source: (Circle Economy, 2025; OECD, 2022; UN-Water, 2024; Friedlingstein et al., 2025; Our World in Data, 2025a). Wastewater share ranges between 56% (2024) and 58% (2022); sinks absorb about half of CO_2 emissions. Flow diagram is schematic.

The accounting rule is the same as in a bath: the level rises whenever the tap runs faster than the drain. The metaphor does not mean that all waste is unmanageable, since lead in petrol and ozone-depleting chemicals were cleared once countries chose to act. It means that where inflow exceeds the drain, accumulation is the default, and the cost of cleaning up rises with the size of the stock.

SECTION 05

Civilizational Decay: Complexity, Fatigue and the Lessons of History

Decay is not a sudden event. It is the slow loss of the capacity to repair, absorb and adapt, which can go unnoticed while output keeps rising.

IN BRIEF

- Complexity pays diminishing returns, and its maintenance bill keeps growing.
- Damage from repeated shocks adds up, and deferred repair turns survivable loads into failures.
- History shows collapse is usually gradual, uneven and multi-causal, with simplification rather than disappearance.

5.1 The rising price of complexity

The anthropologist Joseph Tainter proposed that societies solve problems by becoming more complex: more offices, rules, specialists, infrastructure and information. Each new layer brings benefits, but the benefits per unit of cost shrink as the easy gains are taken. Beyond some point, adding more complexity costs more than it returns, and the society becomes vulnerable to any shock that forces it to spend more than it has (Tainter, 1988). The right panel of Figure 4 shows this as a simple pair of curves: benefit rises and flattens, cost rises steadily, and net benefit peaks and then turns down. It is the same shape as the net-energy cliff and the same logic as a post-peak civilization.

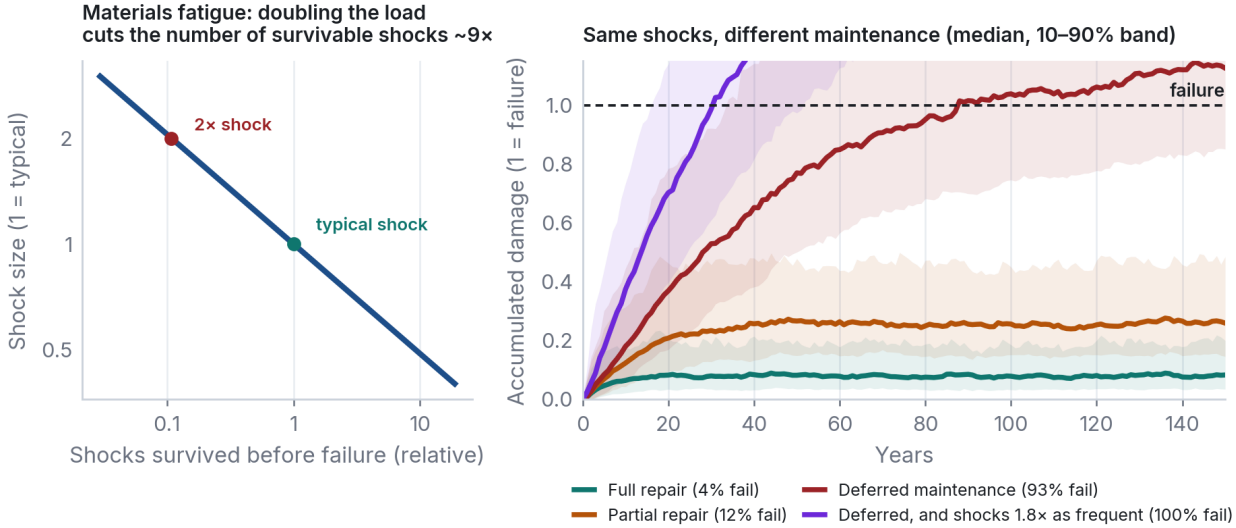
Economists have a cousin of this idea. Mancur Olson argued that long-stable societies accumulate special-interest groups that capture rules and slow adaptation, so growth declines even when nothing external goes wrong (Olson, 1982). Francis Fukuyama described a related process of political decay, in which institutions built for one era cannot reform when conditions change (Fukuyama, 2014). Neither claim is proved. But they point to the same place as the physics: the main danger to a rich, complex society may be its own growing difficulty in maintaining itself.

5.2 Fatigue: damage that adds up

Metal fatigue has a close social analogue, and it is easy to simulate. Figure 7 imagines a structure, which could be a power grid, a health service or a flood defence, hit by random shocks of varying size once a year on average. Each shock adds damage that grows quickly with its size, following the S–N law that engineers use, and each year a fraction of the damage is repaired. With full repair (20% a year), only 4% of 300 simulated structures fail within 150 years. With partial repair (8%), 12% fail. With deferred maintenance (2%), 93% fail. If shocks are also 1.8 times as frequent, every structure fails, with a median failure at year 30. Nothing differs between runs except how much is repaired.

Figure 7 Same shocks, different maintenance: deferred repair turns survivable loads into failures

Left: S–N relation between shock size and shocks survived. Right: accumulated damage under four maintenance regimes



Source: author's simulation (300 runs per case). Damage per shock = s^3.2 ÷ 110, shock size s lognormal; yearly repair of 20%, 8% or 2% of accumulated damage. Illustrative, not calibrated to any single system.

The social translation is direct. A country that postpones bridge repairs, underfunds health systems or lets river embankments age is not failing today. It is reducing the margin that will matter when the next bad flood, heatwave or financial shock arrives. Because damage from many moderate shocks is cumulative, the failure that eventually occurs will be blamed on the last shock rather than the missing maintenance.

5.3 Eroded buffers and tighter coupling

Efficiency and resilience pull in opposite directions. Just-in-time supply chains, thin inventories, single-source suppliers and highly leveraged balance sheets cut costs in normal times and remove the slack that absorbs shocks. Sociologist Charles Perrow argued that systems that are both complex and tightly coupled will have 'normal accidents', since no one can foresee every interaction (Perrow, 1984). In ecology, Holling showed that systems optimised for steady output can lose the ability to absorb disturbance (Holling, 1973). And May's mathematical result that, other things equal, larger and more densely connected random systems are *less* stable warns against assuming that more links always mean more safety (May, 1972).

Formal models point the same way. Bettencourt and colleagues showed that if innovation grows faster than a city's size, growth would reach infinity in finite time, so cities and economies must keep renewing their ideas or stall (Bettencourt et al., 2007). The HANDY model of Motesharrei and colleagues found that collapse can follow from resource depletion combined with steep inequality, and that it is avoidable with more equal use and slower depletion (Motesharrei et al., 2014). Herrington compared the World3 'limits to growth' model with fifty years of data and found that observed trends track its business-as-usual and comprehensive-technology scenarios more closely than its others, although critics note the model's heavy simplification (Herrington, 2021). These are warnings about trajectories, not predictions.

5.4 What history says about collapse

Historical collapses give a reality check. Table 5 summarises five. Four points stand out. First, collapse is typically **simplification, not disappearance**: cities shrink, states fragment, long-distance trade and literacy decline, but people and cultures continue (Tainter, 1988). Second, causes are **compounding**: Cline's account of the Late Bronze Age describes drought, earthquakes, famine, invasions and trade disruption interacting in a 'perfect storm' (Cline, 2014). Third, pace **varies** from a few years to centuries, and the experience is uneven: some groups lose, others gain, as Scott argues for early states whose subjects sometimes escaped to a lighter life (Scott, 2017). Fourth, collapse is not equally likely: adaptive, diversified societies often reorganise rather than fall (Cumming and Peterson, 2017).

Table 5 Five historical cases of decline: speed, stressors and what continued

Case	Approximate dates	Pace	Stressors scholars emphasise	What continued
Late Bronze Age eastern Mediterranean	c. 1200–1150 BCE	Decades	Drought, earthquakes, invasions, broken trade networks, overstretched palace economies	Smaller states; some cities; iron-age reorganisation
Classic Maya (southern lowlands)	c. 800–950 CE	Generations	Long droughts, warfare, political fragmentation, heavy land use	People moved north and to coasts; Maya culture persists
Western Roman Empire	c. 395–476 CE	About a century	Fiscal strain, civil wars, military pressure, plagues, lost provinces	Eastern empire; successor kingdoms; the Church and law
Indus (Harappan) cities	c. 1900–1700 BCE	Centuries	Shifting monsoon and rivers, falling trade, deurbanisation	Smaller villages and regional cultures
Soviet Union	1989–1991	Two to three years	Economic stagnation, loss of legitimacy, reform shocks, nationalisms	Russia and 14 other states; much of the physical infrastructure

Author's summary of scholarly accounts (Tainter, 1988; Cline, 2014; Scott, 2017; Cumming and Peterson, 2017). Dates are approximate and debated by specialists.

What is different today? Three things raise the stakes: **global coupling**, so that a failure can propagate across continents rather than a region; **fossil-fuelled scale**, so the population dependent on a working system is eight billion rather than millions; and **planetary-scale shocks**, since the climate and biosphere are now part of the system being stressed. Three things lower them: **knowledge**, since societies today can see the threat, model it and share solutions; **wealth and technology** that earlier societies lacked; and **institutions of cooperation**, however imperfect. Whether the first three outweigh the second three is the open question of the century.

SECTION 06

The West as a Test Case

The Western world industrialized first, carries the oldest infrastructure and the most complex institutions, and has the clearest signs of slowing, ageing and strain. It also remains rich, healthy and capable.

IN BRIEF

- Western growth per person fell from 3.5% a year in the 1960s to 1.4% in the 2010s, and its share of world output fell from 46% to 32% in 34 years.
- Debt, interest costs, maintenance backlogs, ageing and low institutional trust all point toward institutional strain.
- Relative decline is clear, institutional decay is partly supported, and systemic collapse is not supported by current data.

6.1 What decay and collapse would mean here

The West here means 22 high-income economies of North America, Western Europe, Australia and New Zealand, studied because they industrialized first, not because they are uniquely at fault or at risk. It helps to separate three claims that are often run together. **Relative decline** means a falling share of world output and influence, which is an arithmetic result of other regions catching up. **Institutional decay** means a falling ability to maintain infrastructure, balance budgets, reform rules and keep public trust. **Systemic collapse** means failure of essential services such as power, water, food, health care and order. The first is happening and is largely benign for the world. The second is partly supported by the data below. The third is not supported by any current indicator. The question is whether the second could, through shocks and feedbacks, reach the third.

6.2 A slowing and shrinking share of the world economy

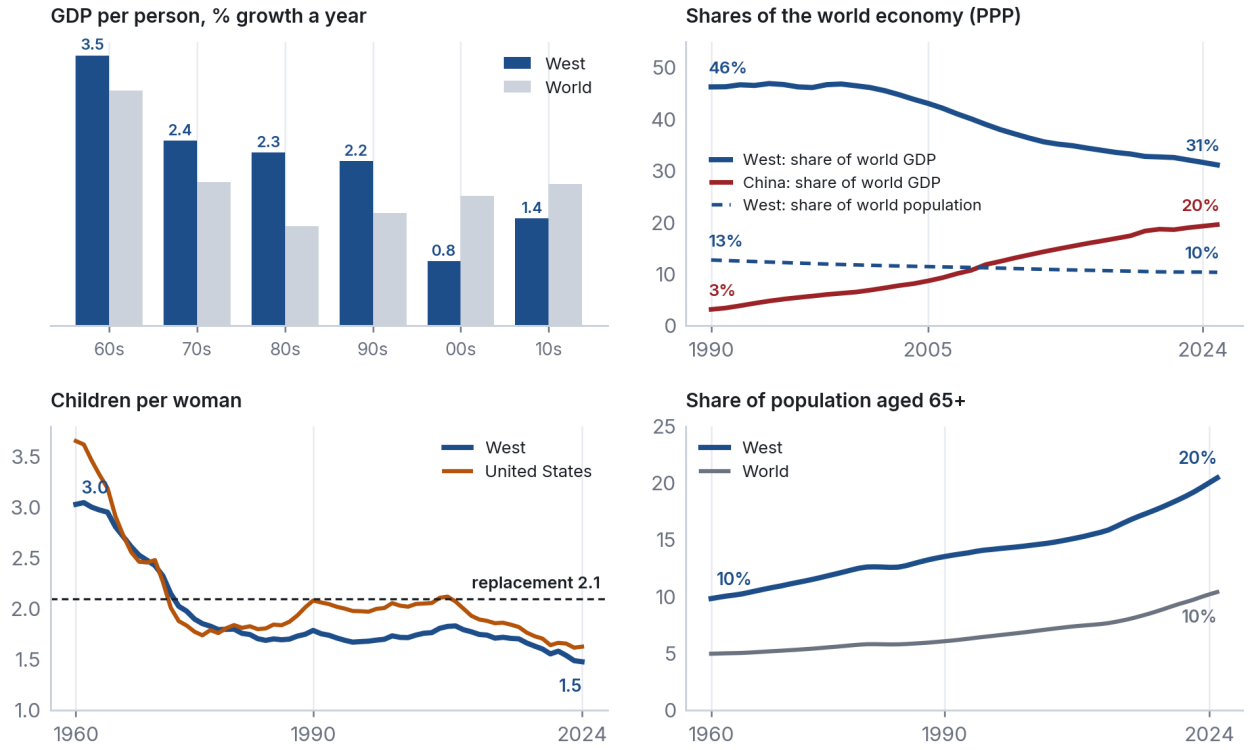
The 22 Western economies in this study grew income per person 3.5% a year in the 1960s, 2.3% in the 1980s, 0.8% in the 2000s, which includes the 2008 financial crisis, and 1.4% in the 2010s, below the world average in both of the last two decades (Figure 8, top left) (World Bank, 2026). Their share of world output at purchasing-power parity fell from 46% in 1990 to 32% in 2024, while China's rose from 3% to 19%. None of this is a disaster in itself, since poorer countries growing faster is what the postwar order aimed for, but the West can no longer expect world growth to carry its budgets.

6.3 Ageing and low fertility

Fertility in the West has fallen from 3.0 children per woman in 1960 to 1.5 in 2023, well below the replacement level of 2.1, and the United States is at 1.6. The share of the population aged 65 or over is now 20%, double the world average of 10% and twice its own level in 1960 (Figure 8, bottom). Fewer workers per retiree raises pension and health costs and leaves fewer people to maintain a very large capital stock.

Figure 8 Growth per person has slowed below the world average, shares of world output have shrunk, and the population is ageing

Growth per person by decade; shares of world output and population; children per woman; share aged 65+



Source: World Bank WDI, author's calculations (World Bank, 2026). 'West' = 22 economies (US, Canada, Western Europe, Australia, New Zealand), population-weighted. GDP per person is in constant US dollars; shares at purchasing-power parity.

6.4 The balance sheet: debt, interest and investment gaps

Government debt has climbed in most large Western economies. According to the IMF, general-government gross debt rose from about 65% of GDP in the US in 2007 to about 124% in 2025, from 43% to 102% in the UK and from 66% to 116% in France; Japan is above 200% and Italy above 130% (Figure 9, top left) (IMF, 2025). The US Congressional Budget Office projects net interest costs of about $1.0 trillion in 2026, or 3.3% of GDP, exceeding defence spending in every year of its outlook and reaching 4.6% of GDP by 2036 (CBO, 2026). Interest payments took 20% of US government revenue in 2024, against 11% in 2010 (World Bank, 2026). The Draghi report found that the EU needs €750–800 billion a year of extra investment, with output per worker growing 0.5% a year since 2019 against 1.6% in the US (Draghi, 2024). High debt is not a collapse, but it reduces room to move when the next shock arrives, as in the fatigue model.

6.5 Maintenance backlogs

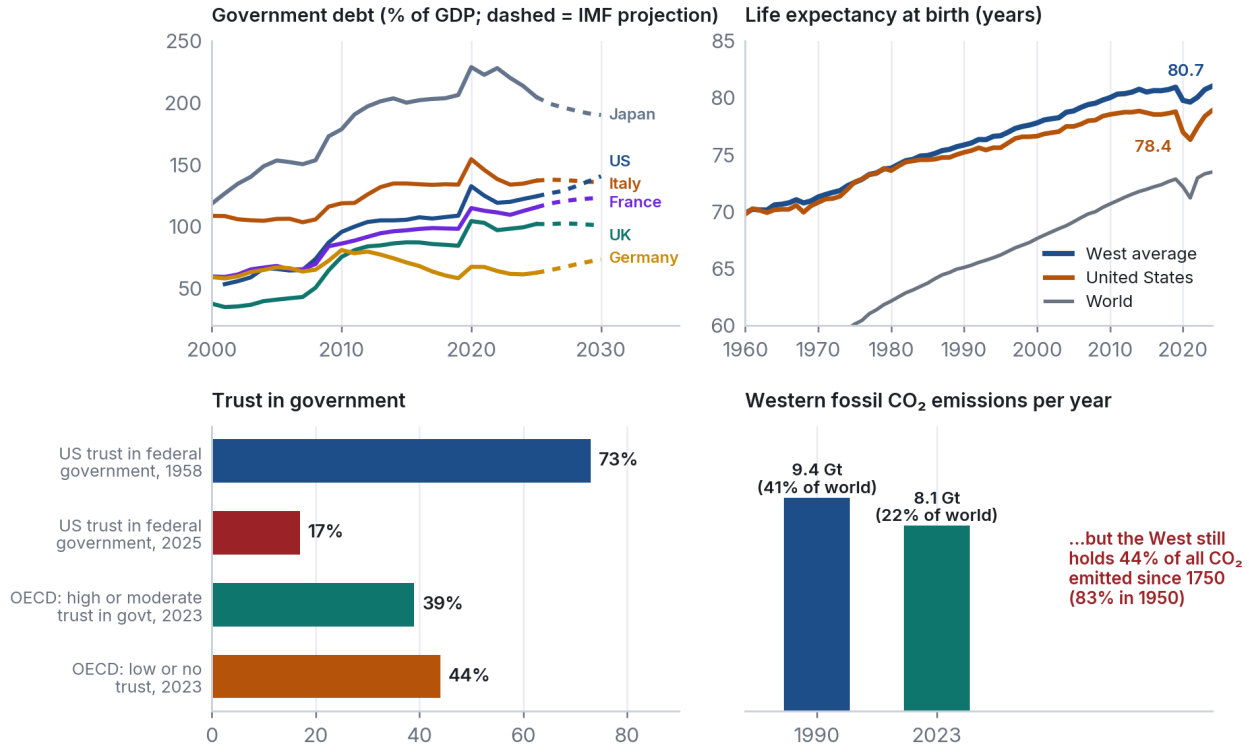
Infrastructure built in the postwar boom is reaching the end of its design life. The American Society of Civil Engineers (ASCE) gave US infrastructure a C in 2025, its highest grade since the report began in 1998, but estimated a $3.7 trillion gap between planned and needed investment from 2024 to 2033, with stormwater and transit graded D (ASCE, 2025). Failures in rich countries are real: the Morandi bridge in Genoa collapsed in 2018 with 43 deaths, and part of Dresden's Carolabrücke fell in 2024 (Wikipedia, 2018; Wikipedia, 2024).

6.6 Institutions, trust and health

Trust is the social glue that lets institutions work, and structural-demographic analysts link its erosion in the US to elite competition and stagnant living standards (Turchin and Korotayev, 2020). In the US, the share who trust the federal government to do the right thing always or most of the time was 73% in 1958 and 17% in September 2025 (Pew Research Center, 2025). Across the OECD, the 2023 survey found 39% with high or moderate trust in national government and 44% with low or no trust (OECD, 2024). Health outcomes show a similar split. Western life expectancy averaged 80.7 years in 2023, but US life expectancy was 78.4, a gap of 2.3 years against 0.2 in 1980, a divergence researchers link to overdoses, suicides, heart disease and obesity (Woolf and Schoomaker, 2019) (Figure 9, top right).

Figure 9 Debt, falling trust and a widening US health gap show institutional strain; emissions at home have fallen but the carbon legacy remains

Government debt; life expectancy; trust in government; Western fossil CO_2



Source: IMF, Pew, OECD, WDI, Our World in Data (IMF, 2025; Pew Research Center, 2025; OECD, 2024; World Bank, 2026; Our World in Data, 2025). Dashed debt lines are IMF projections. West CO_2 is territorial; consumption-based emissions in 2022 were 9.9 Gt against 8.5 Gt territorial.

6.7 The outsourced excretion

The metabolic ledger also tells a mixed story. Territorial fossil CO_2 in the West fell from 9.4 Gt in 1990 to 8.1 Gt in 2023, and its share of the world total fell from 41% to 22% (Figure 9, bottom right). But Western consumption-based emissions, which count carbon embedded in imports, were 9.9 Gt in 2022, 16% above territorial emissions, and the West holds 44% of all CO_2 ever emitted. Part of the Western clean-up has therefore been an export of production and its residuals: an excretion deficit moved, not cleared.

6.8 Strengths and the verdict

Against these signs stand large strengths: high incomes, deep capital markets, a strong research base and a record of reform after crises. Table 6 sets out both sides.

Table 6 The Western scorecard: signs of decay and signs of resilience

Domain	Signs of decay	Signs of resilience	Net reading
Growth	1.4% a year in the 2010s, below world	Income per person ~$53,000, highest of large regions	Slowing, not shrinking
Demography	Fertility 1.5; 20% aged 65+	Immigration; high health and productivity of older workers	Strain on budgets
Fiscal	Debt ~100–125% of GDP in US, UK, France; interest ≈ 20% of US revenue	Deep bond markets; reserve currencies; taxable capacity	Reduced room to move
Infrastructure	$3.7 trillion US gap; ageing bridges and grids	Recent investment lifted US grade to C	Fatigue risk, repairable
Trust	17% US trust; 44% OECD low trust	Rule of law; free press; elections	Weak but functioning
Health	US life expectancy 78.4, 2.3 yrs below peers	West average 80.7 years; advanced medicine	US outlier

Author's assessment from the data in this section.

PATHWAY

A plausible chain from decay to breakdown (not a forecast)

- **1. A shock meets thin buffers.** A drought, heatwave, financial panic or pandemic arrives when fiscal room, spare capacity and goodwill are low.
- **2. Service failures accumulate.** Deferred maintenance shows up as outages and breakdowns in power, water, transport and health care.
- **3. Trust and cohesion erode.** Low trust makes emergency measures and coordination harder, so the response is slower.
- **4. External dependencies bite.** Reliance on imported energy, chips, minerals and capital turns a distant shock into a domestic one (Tooze, 2022).
- **5. Capacity erodes further.** Emergency spending crowds out maintenance, so the next shock lands on a weaker base (Section 8).

The verdict is layered. **Relative decline** is clear and mostly benign; **institutional decay** is partly supported and reversible; **systemic collapse** is not supported by current indicators. The risk lies in the combination: ageing and fiscal strain narrow the West's ability to absorb shocks just as climate, supply-chain and geopolitical pressures grow (World Economic Forum, 2026).

SECTION 07

Tipping Points and Cascades: When Decay Turns Abrupt

Stress can build for decades with little visible change and then produce a sudden shift. Tight coupling between systems makes that shift spread.

IN BRIEF

- Systems with feedback can have two stable states, and the path back from the bad one needs far less stress than the path in.
- Rising memory and variability in a system can warn that a threshold is near.
- Coupled networks fail in cascades: a collapse that one network would survive can destroy two linked ones.

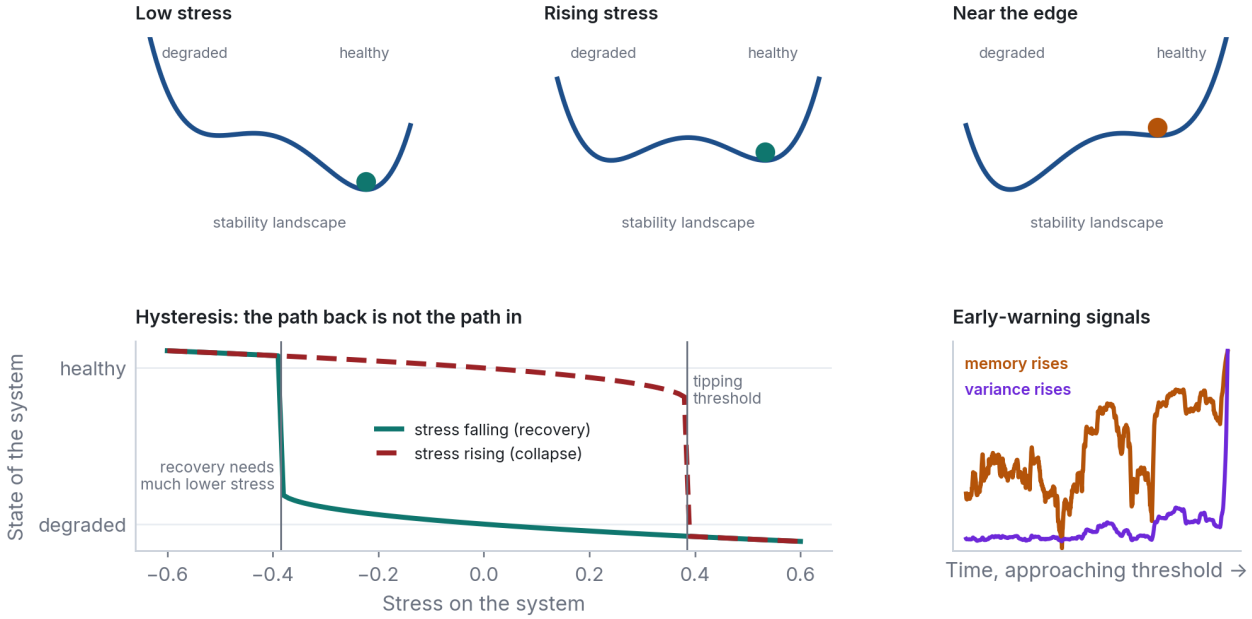
7.1 Gradual stress, abrupt change

Imagine a ball in a landscape of valleys (Figure 10, top). In a healthy state the ball rests in a deep valley and small knocks are absorbed. As stress rises, the valley becomes shallower and a second valley, the degraded state, deepens. Eventually the healthy valley disappears and the ball rolls into the other one. This is a *bifurcation*. The central plot shows its most important consequence, *hysteresis*: the stress at which the system collapses (about +0.38 in the model's units) is far higher than the stress at which it recovers (about −0.38). Once the system has tipped, simply returning to the stress that existed before the collapse is not enough. Lakes that have tipped into an algal state stay there until nutrient loads fall far below the level that caused the shift (Scheffer et al., 2001).

The Earth system contains such elements. A synthesis of the evidence identified sixteen candidate tipping elements and found that warming of 1.5 °C is already within the range at which several could pass their thresholds, including the Greenland and West Antarctic ice sheets and warm-water coral reefs (Armstrong McKay et al., 2022). In October 2025 the Global Tipping Points Report concluded that warm-water coral reefs have passed their thermal tipping point and that widespread mortality is underway, since marine heatwaves hit about 80% of reefs in 2024 (Lenton et al., 2025). The planetary-boundaries framework (Rockström et al., 2009) now finds seven of nine boundaries transgressed in the 2026 Planetary Health Check, up from six in the 2023 assessment (Planetary Boundaries Science Lab, 2026; Richardson et al., 2023). These are measured facts, not model results. Whether the main ice sheets or ocean circulation will tip, and when, remains uncertain.

Figure 10 A system near a tipping point shows a shallower healthy state, a one-way jump and a long road back

Landscapes, hysteresis loop and early-warning signals in a simple model



Source: author's simulation of the fold normal form $dx/dt = r + x - x^3$, with r = −stress. Top: stability landscapes for r = +0.3, 0 and −0.3; bottom left: equilibria as stress is raised and lowered; bottom right: autocorrelation and variance in a noisy run approaching the threshold (Scheffer et al., 2009).

7.2 Early-warning signals

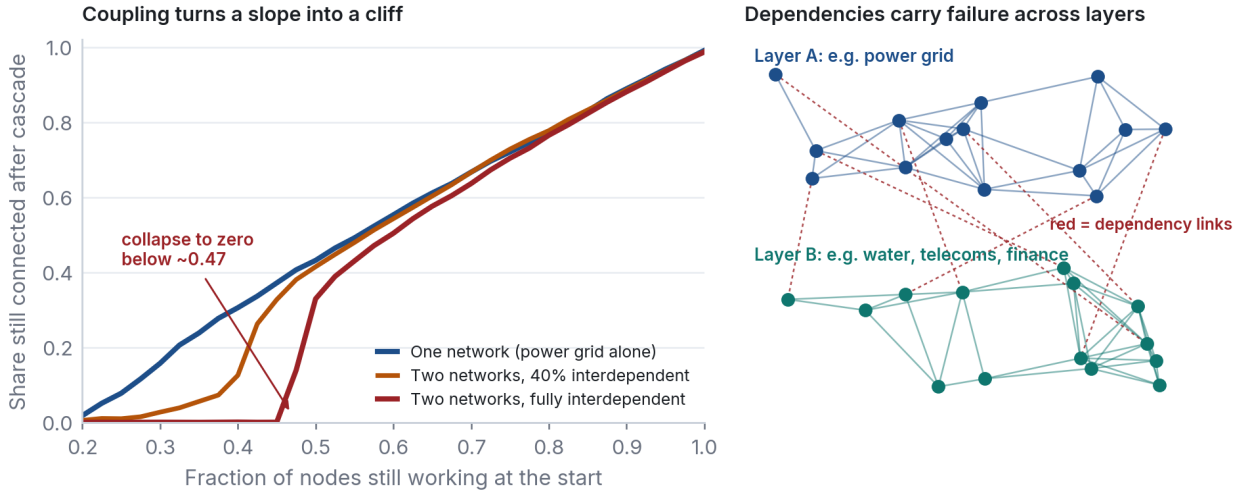
Near a tipping point the valley is shallow and recovery from small knocks takes longer. This *critical slowing down* leaves two statistical fingerprints, rising memory (each state resembles the previous one) and rising variance, both visible in the bottom right of Figure 10, and they have been found in lakes, in laboratory experiments and in climate records (Scheffer et al., 2009). If similar signals could be found in societies, they would give a warning. Candidates are slower recovery after shocks, longer supply-chain disruptions, rising volatility in prices or political outcomes, and growing gaps between stressed and unstressed regions. No such signal has yet been validated for societies, and a lack of warning must not be read as safety. This is a research priority in Section 10.

7.3 Cascading failure in coupled networks

Modern societies depend on layered networks: power grids, water systems, telecommunications, finance and supply chains. Each is more efficient when it relies on the others. Power stations need water and data networks, which in turn need power. Buldyrev and colleagues showed that such interdependence changes how networks break. A single network loses connectivity gradually as nodes fail. Two networks in which each node depends on a node in the other lose it in an abrupt, first-order transition, because every failure in one layer triggers failures in the other, which feed back (Buldyrev et al., 2010). Figure 11 reproduces the result in a simulation of two random networks of 1,500 nodes each: with full dependency, everything is lost once fewer than about 47% of nodes survive the initial shock, while a single network degrades smoothly. With only 40% of nodes dependent, the cliff is lower and later.

Figure 11 **Interdependence turns a slope into a cliff: a single network degrades gradually while two coupled networks collapse abruptly**

Left: share of nodes still connected after the cascade. Right: schematic of dependency links between layers



Source: author's simulation, 1,500-node Erdős–Rényi networks with mean degree 5, mean of four runs per point, after Buldyrev et al. (Buldyrev et al., 2010).

The practical lessons match what engineers and risk analysts have long said: limit unnecessary dependencies, keep firebreaks and redundancy, and protect the hubs (Helbing, 2013). They also tie to the earlier sections: deferred maintenance (Section 5) lowers the fraction of nodes that survive a shock, and a lower starting fraction moves the system closer to the cliff. Network failure is where fatigue meets coupling.

7.4 A live test: the 2020s

The 2020s have offered a live test. A pandemic, supply-chain breakdowns, an energy shock after the invasion of Ukraine, inflation and record heat followed within about three years, which led commentators to adopt the word polycrisis (Tooze, 2022). Most systems bent without breaking, which is evidence of real resilience, but recovery was uneven and several indicators in Section 8 remain elevated. A system that survives such a run has not shown that it can survive a longer or harder one, since fatigue and hysteresis make the last test a poor guide to the next.

SECTION 08

Polycrisis Unravelling: The Feedback Loop

A polycrisis is not many separate problems. It is a loop in which each stress adds to the others, and the loop gain decides whether shocks fade or grow.

IN BRIEF

- Climate, pollution, poverty and conflict each feed the other three.
- If a loop passes on a fraction g of each stress, any shock is amplified by 1 ÷ (1 − g), and above 1 it runs away.
- Capacity, meaning buffers, institutions and income, is what keeps the gain below 1, and it can itself be eroded.

8.1 The four-way loop

The term polycrisis was popularised in 2022 (Tooze, 2022). Four stresses recur in the research on it: climate change, pollution, poverty and conflict (Lawrence et al., 2022; World Economic Forum, 2026). Each is linked to the others by well-documented channels. **Climate** worsens poverty by cutting harvests, destroying assets and raising heat stress; it worsens pollution through wildfire smoke and ozone formation; and by increasing scarcity it can raise the risk of conflict. **Pollution** harms health and labour, and the biomass and coal burned by poor households are significant sources of both carbon and particulates. **Poverty** reduces the ability to adapt, so that each hazard does more damage, and it pushes households toward cheap, dirty fuels and land clearing. **Conflict** destroys infrastructure, diverts budgets, displaces people and blocks the cooperation that climate and pollution policy need. The result is a reinforcing loop (Figure 12, left).

Table 7 gives the current readings. Global temperature in 2025 was 1.43 °C above the pre-industrial level (WMO, 2026); air pollution is linked to 7.9 million deaths a year (Health Effects Institute, 2025); 808–831 million people live below the $3-a-day line (World Bank, 2025); 117.8 million people are forcibly displaced, the first fall in a decade (UNHCR, 2026); and state-based conflicts killed about 154 thousand people in 2025, after a peak of 285 thousand in 2022 and compared with 116 thousand in 2014 (UCDP, 2025). The World Economic Forum's 2026 survey ranked geoeconomic confrontation first among near-term risks (World Economic Forum, 2026). The evidence is mixed. Some indicators, such as displacement and air-pollution deaths, have recently eased; others, such as temperature and CO_2, set records.

Table 7 The state of the four stresses, 2025–26

Stress	Indicator	Latest value	Direction	Source
Climate	Global temperature above pre-industrial	1.43 ± 0.13 °C (2025)	Rising; 2024 record 1.55 °C	WMO
Climate	Fossil CO_2 emissions	38.1 Gt (2025, projected)	Record high	Global Carbon Project
Earth system	Planetary boundaries transgressed	7 of 9 (2026)	All seven worsening	Planetary Health Check
Pollution	Deaths linked to air pollution	7.9 million (2023)	Slightly down from 8.1 m (2021)	State of Global Air
Poverty	People below $3.00 a day	808–831 million (2025)	Falling slowly; 2.3 bn in 1990	World Bank
Poverty	Chronic hunger	About 645 million (2025)	Falling slowly	SOFI
Conflict	Deaths in state-based conflict	About 154,000 (2025)	Peak 285,000 in 2022	UCDP
Conflict	Forcibly displaced people	117.8 million (end 2025)	First fall in a decade	UNHCR

(WMO, 2026; Friedlingstein et al., 2025; Planetary Boundaries Science Lab, 2026; Health Effects Institute, 2025; World Bank, 2025; FAO et al., 2026; UCDP, 2025; UNHCR, 2026).

8.2 Loop gain and amplification

Engineers know the same structure in an amplifier. If a microphone picks up its own loudspeaker, the sound re-enters and is amplified again. If the fraction returned each time, the *loop gain* g, is below 1, the howl dies away; if it reaches 1, it runs away. The mathematics is the same here. Suppose each stress adds its own external driver d and also receives a fraction of every other stress, so the vector of stresses x obeys $x = d + Ax$. Then $x = (I - A)^{-1} d$, and for a single feedback loop with gain g the response to any shock is multiplied by $1 \div (1 - g)$ (Figure 12, right): ×1.4 when g = 0.3, ×2.5 at 0.6, ×5 at 0.8, ×10 at 0.9, and unbounded at 1. The same mathematics lies behind the economic multiplier.

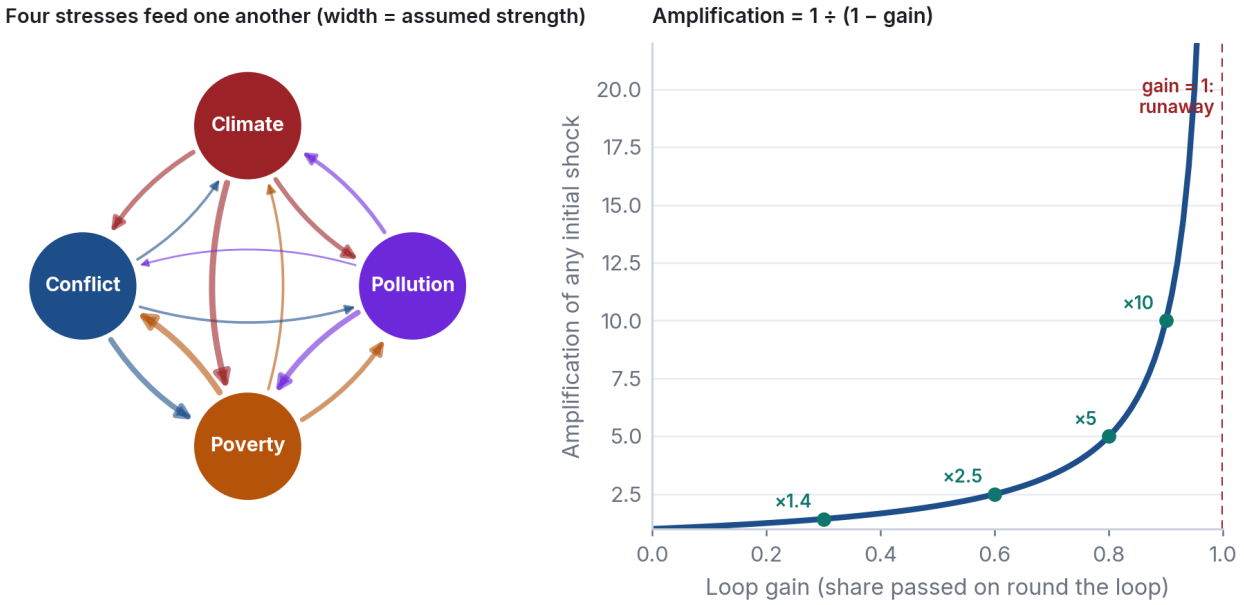
The couplings drawn in Figure 12 are assumptions chosen to be plausible, not estimates. The literature supports the sign of each link, but not a precise magnitude, and the realised gain varies by place and time. Two lessons hold regardless of values. First, **amplification rises faster than the gain**: the difference between g = 0.8 and g = 0.9 is a doubling of the response. Second, a loop that looks harmless when each link is weak can be dangerous when four links reinforce each other. A world in which every individual relation is modest can still have a large combined gain.

8.3 Capacity decides the gain

What keeps the gain below 1 is capacity: the ability of institutions, infrastructure, ecosystems and households to absorb stress. A country with reliable power, strong public health and fiscal room can take a drought or a pandemic without it spilling into unrest or collapse. A country without them cannot. In the author's earlier cross-country work, the average correlation between ten vulnerabilities fell from 0.51 to 0.24 once income was held constant, meaning that poverty and low capacity explain much of why the stresses travel together (Hughes, 2026b). Where capacity is high, the same hazard has smaller effects; where it is low, a hazard amplifies.

Figure 12 Four stresses feed one another; the amplification of any shock grows as 1 ÷ (1 – gain) and becomes unbounded at a gain of 1

Left: the assumed feedback structure. Right: amplification as a function of loop gain



Source: author's model. Coupling strengths are illustrative assumptions with signs supported by the literature (Lawrence et al., 2022). The spectral radius of the assumed coupling matrix is 0.64 before the effect of capacity.

The deepest risk is a **second loop**: stress erodes capacity, and low capacity raises the gain. Repeated shocks deplete reserves, fiscal space and trust, so the next shock lands on a weaker system and passes on more. This is the social version of the fatigue model of Section 5, and it is what could turn a prolonged polycrisis into an unravelling. It is also where policy can act, because capacity can be rebuilt. Section 9 puts both loops into a simple stochastic model.

Unravelling will not be uniform. It is likely to begin where capacity is lowest and shocks are largest: fragile and conflict-affected states, small islands, and cities exposed to heat and floods. Rich regions are buffered but not immune. They are exposed through energy and food prices, supply chains, migration and finance, which is why an unravelling elsewhere is also a risk for the West.

SECTION 09

Polycrisis Apocalypse: Scenarios for This Millennium

The extreme outcome is a persistent, widespread loss of survivability. It needs both rising pressure and eroding capacity, and it is a tail of the distribution, not its centre.

IN BRIEF

- In a stylised model with 400 runs per path, 0% of managed, 43% of prolonged and 95% of severe runs are past the unravelling threshold by 2100.
- Even the best path leaves multi-century legacies in CO_2, sea level and lost species, with world population peaking near 10 billion in the 2080s (United Nations, 2024).
- Expert estimates of global catastrophe range from a few percent to about one in five this century.

9.1 What a Polycrisis Apocalypse would mean

The word *apocalypse* here means a state, not an event: a **persistent, widespread deterioration in human survivability, essential services, ecological conditions and civilizational function**. It is not human extinction. Extinction-level risk comes mainly from other sources, such as engineered pandemics, nuclear war or misaligned artificial intelligence, and expert estimates for climate change as a direct cause are small (Ord, 2020; Kemp et al., 2022). The risk analysed here is of a long, broad decline in which many societies lose the ability to provide power, water, food, health care and order, and the loss feeds on itself. Plausible markers include a sustained fall in global life expectancy over decades, the failure of food or energy systems across several regions at once, and mass displacement that exceeds the absorptive capacity of receiving regions.

9.2 Three paths to 2100

Table 8 defines three paths, and Figure 13 shows a stylised model of each. The model tracks four stresses, climate, pollution, poverty and conflict, that feed one another through the matrix of Figure 12. External pressure rises to 2025 and then follows each path. Capacity, which both relaxes stress and weakens coupling, is built toward a target and is *eroded when stress is high*, the second loop. Random shocks, more likely when climate stress is high, hit the stresses at random. Each path was run 400 times with random variation in shocks and in key parameters. The model is a thinking tool, not a forecast: its parameters are chosen to reproduce a rise in stress to 2025, and not fitted to data.

Three results stand out. First, all three paths are similar until about 2045, since stress is driven mostly by past accumulation, and they diverge afterwards. Second, the managed path peaks around 2045 at about 0.61 and then falls, because capacity is built quickly enough to cut the loop gain. Third, the difference between the prolonged and severe paths is not in the central case, where both worsen, but in the tail: 43% of prolonged runs cross the threshold by 2100 against 95% of severe ones, and in the severe path the median itself crosses around 2073. The conclusion

Table 8 Three paths to 2100: assumptions and stylised model results

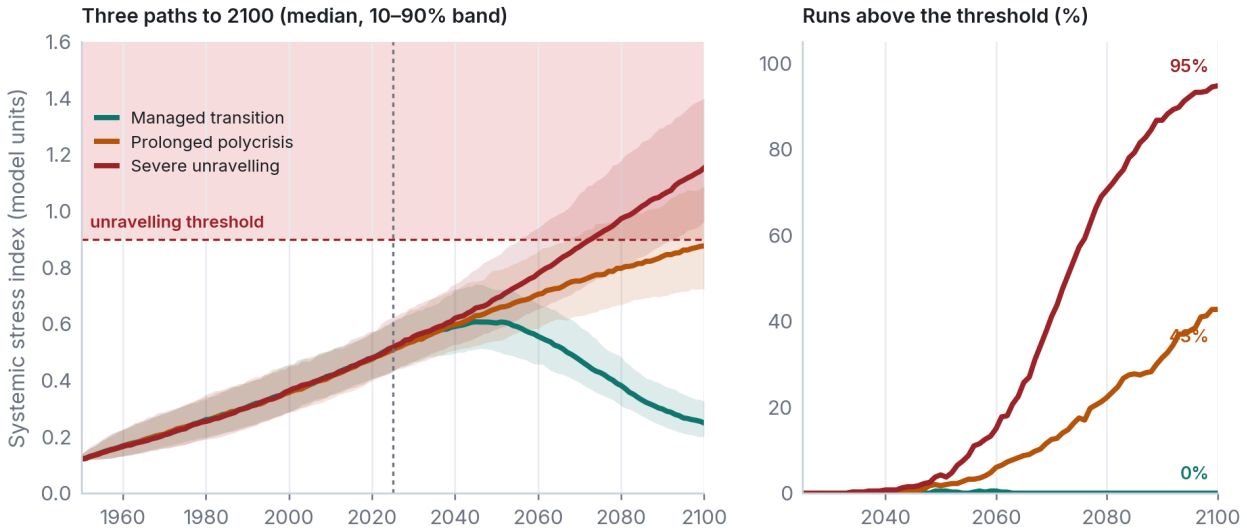
Item	Managed transition	Prolonged polycrisis	Severe unravelling
Climate	Emissions fall to net zero by about 2070; warming peaks near 2 °C	Current policies; about 2.8 °C by 2100 (UNEP, 2025)	High emissions; 3.5–4 °C or more by 2100 (IPCC, 2021)
Capacity (2100)	Rises: 1.10 (2025: 0.83)	Slowly erodes: 0.72	Erodes: 0.58
Shocks and coupling	Fewer shocks; coupling cut by redundancy and cooperation	Frequent shocks; coupling unchanged	More frequent shocks; coupling rises as capacity falls
Median stress, 2050	0.60	0.65	0.69
Median stress, 2100	0.25	0.88	1.15
Runs past threshold, 2075	0%	18%	57%
Runs past threshold, 2100	0%	43%	95%

Author's model; scenario climate assumptions from (UNEP, 2025; IPCC, 2021). Threshold = stress index 0.9 (about 1.8 times its 2025 level); stress index 0.51 in 2025. Results depend on the parameters and are illustrative.

is a sensitivity, not a probability: whether a Polycrisis Apocalypse occurs depends less on the pressure itself than on whether capacity is allowed to erode.

Figure 13 Three stylised paths diverge after 2045; a prolonged polycrisis risks crossing the threshold in a large minority of runs

Left: median and 10–90% band of the stress index, 1950–2100. Right: share of runs above the threshold



Source: author's stylised stochastic model, 400 runs per path. Systemic stress index in model units, 2025 = 0.51; unravelling threshold = 0.9. Not a forecast.

9.3 Beyond 2100: the millennial tail

The question for this millennium is not only what happens by 2100. Some of what has been done cannot be undone on any human timescale (Table 9). A substantial share of emitted CO_2 stays in the air for many centuries, and the resulting warming is effectively irreversible for at least a thousand years (Archer et al., 2009; Solomon et al., 2009). Sea level responds slowly: the IPCC estimates two-thousand-year commitments of 2–3 m at 1.5 °C of warming, 2–6 m at 2 °C and 19–22 m at 5 °C (Fox-Kemper et al., 2021). Extinct lineages do not return, and long-lived radioactive wastes remain hazardous for tens of millennia. A managed transition still leaves these. They define the baseline against which later societies must adapt.

Table 9 Legacies that outlast any scenario

Legacy	Persistence	Evidence
Carbon dioxide in the atmosphere	Centuries to many millennia	A substantial share remains for many centuries (Archer et al., 2009)
Warming caused by CO_2	At least about 1,000 years	Irreversible on that scale after emissions stop (Solomon et al., 2009)
Sea-level rise	Millennia	2–3 m at 1.5 °C, 2–6 m at 2 °C, 19–22 m at 5 °C over 2,000 years (Fox-Kemper et al., 2021)
Species extinction	Permanent	Lost lineages cannot be restored; coral reefs already past their tipping point (Lenton et al., 2025)
Long-lived radioactive waste	Tens of millennia	Plutonium-239 has a half-life of about 24,000 years
Institutions and knowledge	Decades to centuries to rebuild	Depends on what survives simplification (Tainter, 1988)

As cited in the table. Persistence statements are summaries of the cited sources.

9.4 Probability and humility

How likely is a severe path? Honest answers vary. In the 2022 forecasting tournament, experts judged the chance of a catastrophe that kills at least a tenth of the world's population within five years at about 20% before 2100, and superforecasters at about 9%, and the gap illustrates how uncertain such estimates are (Forecasting Research Institute, 2023). Ord put total existential risk this century at about one in six, but mostly from other sources than climate or pollution (Ord, 2020). Researchers who study climate tails argue that the worst cases are under-studied relative to their consequences (Kemp et al., 2022; Steffen et al., 2018). This report takes no position on a single number. It asks planners to treat the downside tail as a design requirement: the right response to a low-probability, high-consequence risk is not to predict it but to reduce exposure to it.

SECTION 10

Slowing the Decay: Resilience and a Test Programme

The loop has three handles: reduce the pressure, rebuild the capacity and cut the coupling. A thesis about decay should also say what evidence would prove it wrong.

IN BRIEF

- Each physical foundation implies a practical design principle, from maintenance budgets to firebreaks.
- Polycrisis Resilience is the countervailing approach: lower the vulnerability that lets stresses compound.
- Six tests over 2026–2035 would support or refute the post-peak reading.

10.1 Three handles on the loop

The feedback model of Section 8 shows where intervention works. **Reduce the drivers**: cut emissions, pollution and exposure so that less stress enters the loop. **Raise capacity**: build buffers, repair infrastructure, strengthen institutions and expand adaptation, so that stress relaxes faster. **Cut the coupling**: add redundancy, modularity and cooperation so that a failure in one system passes on less to the others. The evidence for the second is strong where it has been tested: the Global Commission on Adaptation reported returns of about 2:1 to 10:1 for resilience investments, and the World Bank found that resilient infrastructure in low- and middle-income countries yields about $4 of benefit per $1 invested (Global Commission on Adaptation, 2019; Hallegatte et al., 2019).

10.2 Design principles from the physics

The five foundations of Section 2 each suggest a principle. Table 10 pairs them with practical examples.

The cheapest unit of resilience is usually bought *before* the shock, and because of hysteresis a system pushed past its threshold may need far more effort to restore than it would have taken to protect.

10.3 Six tests of the post-peak thesis

A framework that cannot be falsified is only a story. Table 11 sets out six tests, each with a result that would support the thesis and one that would refute it.

Table 10 From physical foundation to design principle

Foundation	Principle	What it looks like in practice
Thermodynamics	Lower the energy and material cost per unit of well-being	Efficiency, electrification and clean energy; avoiding waste of useful energy
Metabolism	Build 'excretory organs' that match the flow of residuals	Wastewater treatment, circular materials, carbon removal, pollution standards
Materials fatigue	Fund maintenance first and track the backlog	Asset-management budgets, inspection regimes, reserve funds for repairs
Nonlinear dynamics	Stay a safe distance from thresholds and monitor early-warning signals	Safety margins, limits on drawdown, monitoring of recovery times
Network science	Cut unnecessary coupling and protect hubs	Redundancy, modular grids, diverse suppliers, circuit-breakers

Author's synthesis.

Table 11 Six tests of the post-peak thesis, 2026–2035

Test	Supports the thesis if…	Refutes the thesis if…	Data
1 Yield per unit of liability	Gains per unit of CO_2 and material throughput keep falling	Absolute decoupling: output and well-being rise while emissions and material use fall	Maddison/WDI, GCP, UNEP IRP
2 Net energy	Final-stage EROI of the energy system keeps falling	Renewables and efficiency raise final-stage returns	Energy-balance studies
3 Maintenance	Infrastructure grades and backlog worsen; maintenance share of spending rises	Backlogs shrink and grades rise	ASCE, national asset registers
4 Early warning	Recovery times after shocks lengthen; variance and persistence rise	Recovery times shorten	Trade, price and outage series
5 Coupling	Co-occurring failures become more frequent	Failures stay isolated; redundancy grows	Loss databases, grid and supply-chain data
6 Capacity	Western trust, fiscal room and health keep falling	They stabilise or recover	OECD, Pew, IMF, WPP

Author's proposal.

10.4 Conclusion

The evidence supports a measured version of the thesis. The years from 1950 to 2025 were probably the period of highest net yield from industrial civilization: the best decade for most gains was long ago, while the liabilities have grown every decade. Slowing returns, accumulating residuals, deferred maintenance and thinner buffers are visible, most clearly in the oldest societies.

The evidence does not support fatalism, since collapse is not the default. The model shows that pressure alone does not unravel a society as long as capacity is maintained, and that capacity can be rebuilt. The central question of the coming decades is therefore not whether output will grow, but whether civilization will keep the systems that turn output into lasting well-being in better repair before thresholds are crossed.

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About the author. Hunter Hughes is the Founder of H Heuristics, an independent research practice on polycrisis, resilience and development.

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About this report. This report proposes and tests a framework in which the years 1950–2025 are a candidate peak of net civilizational yield, followed by diminishing returns, decay and a possible polycrisis unravelling. It draws on thermodynamics, metabolism, materials science, nonlinear dynamics and network science, with the Western world as a test case.

Disclaimer. The views expressed are the author's. The simulations are stylised illustrations, not forecasts, and scenario results depend on assumed parameters. Country figures summarise public indicators and are not ratings of governments or advice on investment. Corrections are welcome.